

Evaluating Cross-Validation Reliability in Kaggle Tabular Binary Classification Competitions Using Bayesian Melding

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Abstract—Cross-validation is one of the most widely used methods for estimating machine learning model performance before deployment or external evaluation. In competitive machine learning platforms such as Kaggle, practitioners commonly rely on cross-validation scores and public leaderboard feedback to guide model development, while final model performance is determined using a hidden private leaderboard. This creates an important challenge in determining whether cross-validation estimates reliably predict true generalisation performance.

This study investigates the reliability of cross-validation in Kaggle tabular binary classification competitions using an empirical Bayesian melding framework. Three completed Kaggle competitions were analysed: Santander Customer Transaction Prediction, Porto Seguro’s Safe Driver Prediction, and Homesite Quote Conversion. For each competition, 1,000 controlled model configurations were trained and evaluated using stratified 5-fold cross-validation, resulting in 3,000 cross-validation observations. Due to Kaggle leaderboard submission constraints, a representative subset of 100 models per competition was selected across model families and performance ranges, producing 300 models with cross-validation, public leaderboard, and private leaderboard scores.

The results show very strong alignment between cross-validation AUC and private leaderboard AUC across all competitions, with within-competition centred Pearson correlation of 0.9958. Public leaderboard scores were also highly correlated with private leaderboard scores, with centred Pearson correlation of 0.9988. Bayesian melding was used to treat cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. The estimated global cross-validation bias was small at 0.00042 AUC, while public leaderboard bias was 0.00057 AUC. These findings suggest that, when carefully designed and evaluated across model families and competitions, stratified cross-validation can provide a highly reliable estimate of hidden leaderboard performance in tabular binary classification competitions.

Keywords—cross-validation, Bayesian melding, Kaggle, leaderboard, model validation, binary classification, tabular data, generalisation, machine learning evaluation

I. INTRODUCTION

Machine learning models are only useful when their measured performance reflects how well they generalise to unseen data. In practice, this is difficult to guarantee. During model development, practitioners often estimate model

quality using validation methods such as holdout validation or cross-validation. However, the final performance of a model may differ from validation estimates due to sampling noise, data leakage, model selection bias, distribution shift, or overfitting to validation feedback.

This issue is especially important in competitive machine learning environments such as Kaggle [11]. Kaggle competitions provide participants with training data and a hidden test set. Participants submit predictions for the test set and receive feedback through a public leaderboard. The final ranking is usually determined by a private leaderboard, which is hidden until the competition ends. As a result, practitioners typically make modelling decisions using two imperfect sources of information: local cross-validation scores and public leaderboard feedback [1], [2]. Neither source is guaranteed to perfectly represent private leaderboard performance.

Cross-validation is widely used because it allows the training data to be split into multiple folds, giving a more stable estimate of model performance than a single validation split. However, cross-validation can still be biased [6], [7]. It may overestimate performance when preprocessing or feature selection leaks information across folds, when hyperparameters are repeatedly tuned against the same validation strategy, or when the validation folds do not represent the hidden test distribution. Public leaderboard feedback can introduce another form of bias because users may repeatedly submit models and adjust their approach based on leaderboard movements.

This research studies the reliability of cross-validation in Kaggle tabular binary classification competitions. The focus is not on achieving the best Kaggle rank, but on understanding whether cross-validation scores are trustworthy estimates of hidden private leaderboard performance. This is valuable because many machine learning practitioners use competition results, benchmark scores, and validation pipelines to make decisions about model quality. If cross-validation is reliable, it supports its continued use as a practical model evaluation tool. If it is unreliable, practitioners need stronger validation strategies and more cautious interpretation of leaderboard results.

The study uses three completed Kaggle tabular binary classification competitions: Santander Customer Transaction

Prediction, Porto Seguro’s Safe Driver Prediction, and Homesite Quote Conversion. These competitions were selected because they involve structured tabular data, binary targets, and public/private leaderboard evaluation. For each competition, 1,000 controlled model configurations were trained using six model families: LightGBM, XGBoost, CatBoost, Random Forest, ExtraTrees, and Logistic Regression. Each model was evaluated using stratified 5-fold cross-validation. A representative subset of 100 models per competition was then evaluated using Kaggle public and private leaderboard scores.

To integrate these different evaluation signals, this research adopts a Bayesian melding framework [13]. In this framework, each model has an unobserved true generalisation performance. Cross-validation, public leaderboard, and private leaderboard scores are treated as noisy observations of that latent performance. Cross-validation and public leaderboard scores may include systematic bias, while the private leaderboard is treated as the closest available proxy for true generalisation. This approach allows the study to quantify validation bias, estimate uncertainty, and compare the reliability of different evaluation sources.

The main contribution of this research is an empirical evaluation of cross-validation reliability across 3,000 trained models and 300 leaderboard-scored models in Kaggle tabular binary classification settings. The study provides quantitative evidence on the relationship between cross-validation, public leaderboard, and private leaderboard performance, and demonstrates how Bayesian melding can be used as a principled framework for model validation.

II. RELATED WORK

Machine learning model evaluation has been widely studied in relation to generalisation, validation bias, benchmark reliability, and leaderboard feedback. These issues are central to competitive machine learning, where participants rely on validation scores and leaderboard signals to decide which models to submit and trust.

A major concern in machine learning competitions is leaderboard overfitting. Roelofs et al. conducted a large-scale empirical study of Kaggle competitions and investigated whether repeated submissions lead to overfitting to the public leaderboard [1]. Their work found limited evidence of severe adaptive overfitting in many competitions, although discrepancies between public and private leaderboard scores can still occur when test sets are small, noisy, or not representative. This is directly relevant to the present study because Kaggle users often use public leaderboard feedback as a proxy for final model quality.

Leaderboard design has also been studied as a way to reduce overfitting. Blum and Hardt proposed the Ladder mechanism, which only reveals leaderboard improvements when a submission achieves a statistically meaningful gain over previous submissions [2]. This reduces the amount of information leaked by the public leaderboard. Chaibub Neto

et al. later proposed LadderBoot, a bootstrapped leaderboard mechanism designed to further reduce overfitting in challenge-based competitions [3]. These works show that leaderboard feedback itself is not neutral; the way scores are revealed can influence participant behaviour and evaluation reliability.

The broader problem of adaptive data analysis provides a theoretical foundation for understanding leaderboard overfitting. Dwork et al. showed that repeatedly querying the same validation data can invalidate standard statistical assumptions and lead to overfitting [4]. Their work on reusable holdout methods demonstrates that validation data can be reused more safely if information leakage is controlled [5]. This is closely related to Kaggle competitions because public leaderboard submissions effectively reuse part of the hidden test set as feedback.

Cross-validation bias is another important issue. Varma and Simon showed that using cross-validation both for model selection and performance estimation can produce optimistically biased results [6]. Cawley and Talbot similarly argued that model selection itself can overfit when validation scores are noisy [7]. These findings are important because machine learning practitioners often select models and tune hyperparameters based on cross-validation scores, then treat those same scores as estimates of final performance. This can lead to overconfidence in validation results.

Data leakage is also a common cause of inflated validation performance. Kaufman et al. described leakage as a situation where information from the target or test data is unintentionally used during training or preprocessing [8]. Leakage can occur through feature engineering, preprocessing, or incorrect validation design. In high-dimensional settings, Ambroise and McLachlan showed that feature selection performed before cross-validation can produce strongly biased performance estimates [9]. Although the present study uses consistent preprocessing pipelines to reduce leakage risk, these works show why validation estimates must be interpreted carefully.

Benchmark generalisation research also questions whether fixed test sets fully represent real-world performance. Recht et al. showed that models evaluated on newly collected ImageNet-style data experienced performance drops compared with the original ImageNet benchmark, even though model rankings were relatively stable [12]. This suggests that benchmark scores can preserve relative ordering while still misestimating absolute generalisation performance. A similar question applies to Kaggle competitions: cross-validation may preserve model ranking but still differ from private leaderboard performance.

The initial progress report identified these same themes as the foundation of the research problem: leaderboard overfitting, adaptive data reuse, cross-validation bias, data leakage, and benchmark generalisation. It also defined the primary research direction as evaluating the relationship between cross-validation AUC, public leaderboard AUC, and

private leaderboard AUC in Kaggle tabular classification competitions. The final study extends that direction by scaling the experiment to 3,000 trained models, adding more model families, replacing the originally proposed Facebook competition with Homesite Quote Conversion for a cleaner binary tabular setup, and implementing Bayesian melding as the final validation framework.

Although prior work has studied leaderboard overfitting and validation bias, fewer studies directly compare cross-validation, public leaderboard, and private leaderboard scores across a controlled population of many model configurations and multiple tabular binary classification competitions. This study addresses that gap by combining large-scale empirical experimentation with Bayesian melding to quantify validation reliability and uncertainty.

III. RESEARCH PROBLEM AND QUESTIONS

The main research problem is that model developers often rely on cross-validation scores to estimate final model performance, but the reliability of these estimates is uncertain in competition-based machine learning settings. In Kaggle competitions, the private leaderboard is the final evaluation target, but it is hidden until the competition ends. Practitioners therefore depend on cross-validation and public leaderboard scores when making modelling decisions. If these signals are biased or noisy, they may lead to incorrect conclusions about model quality.

This study investigates whether cross-validation AUC is a reliable predictor of private leaderboard AUC in Kaggle tabular binary classification competitions. It also examines whether public leaderboard scores provide a more reliable estimate of private leaderboard performance than cross-validation, and whether reliability differs across competitions and model families.

The primary research question is:

RQ1: How reliable is stratified 5-fold cross-validation AUC as an estimate of private leaderboard AUC in Kaggle tabular binary classification competitions?

The secondary research questions are:

RQ2: How closely do public leaderboard AUC scores align with private leaderboard AUC scores?

RQ3: Does validation reliability differ across competitions and model families?

RQ4: What systematic bias and noise are associated with cross-validation and public leaderboard estimates?

RQ5: Can Bayesian melding provide a useful framework for integrating cross-validation, public leaderboard, and private leaderboard scores into a unified estimate of latent model performance?

The study makes three main contributions. First, it builds a large controlled experimental population of 3,000 model configurations across three Kaggle competitions. Second, it constructs a representative leaderboard-scored subset of 300 models with complete cross-validation, public leaderboard, and private leaderboard observations. Third, it applies Bayesian melding to model validation reliability by treating each evaluation source as a noisy and potentially biased observation of latent true model performance.

IV. METHODOLOGY AND DATASETS

This study uses an empirical experimental design to evaluate the reliability of cross-validation in Kaggle tabular binary classification competitions. The methodology consists of five stages: competition selection, large-scale model generation, cross-validation evaluation, representative leaderboard scoring, and Bayesian melding analysis.

A. Competition Selection

Three completed Kaggle competitions were selected for this study: Santander Customer Transaction Prediction [14], Porto Seguro's Safe Driver Prediction [15], and Homesite Quote Conversion [16]. All three competitions are tabular binary classification tasks, meaning that each dataset consists of structured features and a binary target variable.

Santander Customer Transaction Prediction asks whether a customer will make a transaction. Porto Seguro's Safe Driver Prediction asks whether a driver will file an insurance claim in the following year. Homesite Quote Conversion asks whether a customer will purchase a quoted insurance plan. These tasks were selected because they represent realistic tabular prediction problems and provide public and private leaderboard scores for submitted predictions.

The original progress report proposed using Santander, Porto Seguro, and Facebook Recruiting IV. During final implementation, Facebook Recruiting IV was replaced with Homesite Quote Conversion. This change was made to maintain a cleaner and more consistent binary tabular classification setup, because Homesite provides a standard train/test/sample-submission structure and a direct binary target. This replacement improved consistency across competitions while preserving the original research aim of studying validation reliability in Kaggle tabular classification settings.

B. Experimental Dataset Design

The final experiment contains two dataset layers. The first layer is the full cross-validation population, consisting of 1,000 trained model configurations per competition. Across three competitions, this produced 3,000 cross-validation observations. This full population was used to analyse model-family performance distributions, cross-validation stability, and the broader search space from which representative leaderboard models were selected.

The second layer is the leaderboard-scored subset. Due to Kaggle submission limits and anti-abuse throttling, it was not practical to obtain public and private leaderboard scores for all 3,000 trained models. Following supervisor approval, a representative subset of 100 models per competition was selected. Across three competitions, this produced 300 models with complete cross-validation, public leaderboard, and private leaderboard observations. This subset was used for validation reliability analysis and Bayesian melding.

The final design therefore uses all 3,000 cross-validation observations for population-level analysis and 300 leaderboard-scored observations for direct comparison between validation estimates and hidden leaderboard performance.

Experimental Workflow

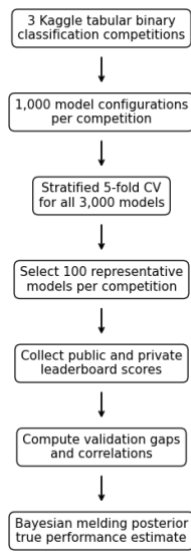


Fig. 1. Experimental workflow showing the construction of the 3,000-model cross-validation population, selection of 300 representative leaderboard-scored models, and Bayesian melding analysis.

C. Model Families and Configuration Generation

Six model families were used: LightGBM [17], XGBoost [18], CatBoost [19], Random Forest [20], ExtraTrees [21], and Logistic Regression. These were selected to represent a range of common tabular modelling approaches, including gradient boosting, bagging-based tree ensembles, extremely randomized trees, and linear classification.

For each competition, 1,000 model configurations were generated using controlled random sampling of hyperparameters. The configuration distribution was kept consistent across competitions: 350 LightGBM models, 350 XGBoost models, 150 CatBoost models, 75 Random Forest / ExtraTrees models, 75 Logistic Regression models.

Random Forest and ExtraTrees were treated as separate implementations but were also interpreted as related bagging-

style ensemble methods. The exact split between Random Forest and ExtraTrees was produced by controlled random sampling. Across the full 3,000-model population, this resulted in 1,050 LightGBM models, 1,050 XGBoost models, 450 CatBoost models, 225 Logistic Regression models, 126 ExtraTrees models, and 99 Random Forest models.

This design intentionally included both high-performing boosting models and simpler or more stable baselines. The purpose was not only to find strong models, but to generate a diverse performance distribution across weak, medium, and strong configurations. This diversity was necessary for analysing whether cross-validation reliability remains stable across different model families and performance levels.

D. Preprocessing and Cross-Validation

All models were evaluated using stratified 5-fold cross-validation. Stratification was used because all three tasks are binary classification problems, and class imbalance can affect validation reliability if folds are not class-balanced.

For tree-based and boosting models, missing values were handled using median imputation. Logistic Regression used median imputation followed by standard scaling. Each model configuration was evaluated using the same preprocessing and validation structure within each competition to ensure fair comparison.

The primary cross-validation metric was area under the receiver operating characteristic curve (ROC-AUC). For Santander and Homesite, Kaggle leaderboard scores were already reported as AUC. Porto Seguro's Safe Driver Prediction uses Normalized Gini as its official leaderboard metric. Since Normalized Gini and AUC are linearly related by:

$$\text{AUC} = (\text{Gini} + 1) / 2$$

the Porto public and private leaderboard scores were converted to AUC before analysis. This allowed all three competitions to be compared using a common AUC-based framework.

E. Representative Leaderboard Subset Selection

For each competition, 100 representative models were selected from the 1,000 trained configurations for Kaggle leaderboard evaluation. The subset was designed to cover both model-family diversity and performance-level diversity.

The selection process grouped models by model family and selected models across low, median, and high cross-validation performance ranges. This avoided selecting only the highest-performing models and reduced the risk of cherry-picking. The final per-competition leaderboard subset consisted of: 25 LightGBM models, 25 XGBoost models, 20 CatBoost models, 15 Logistic Regression models, 15 Random Forest / ExtraTrees models combined.

This produced 100 leaderboard-scored models per competition and 300 leaderboard-scored models in total. In the final combined leaderboard dataset, the model-family distribution was 75 LightGBM, 75 XGBoost, 60 CatBoost, 45 Logistic Regression, 23 ExtraTrees, and 22 Random Forest models. This distribution preserved the intended emphasis on major tabular model families while maintaining representation from simpler and bagging-based models.

F. Leaderboard Score Collection

Each selected model was trained on the full training dataset of its competition and used to generate predictions for the Kaggle test set. The resulting prediction files were submitted to Kaggle to obtain public and private leaderboard scores.

Kaggle submission limits and rate throttling made automated submission of all model configurations impractical. Therefore, leaderboard collection was performed for the approved representative subset of 100 models per competition. Public and private leaderboard scores were manually verified and stored in clean final research datasets.

For each leaderboard-scored model, the following values were recorded: cross-validation AUC mean, cross-validation AUC standard deviation, public leaderboard AUC, private leaderboard AUC, model family, competition, model identifier.

The final leaderboard-scored dataset contained 300 rows, with no missing public or private leaderboard scores.

G. Validation Reliability Metrics

Validation reliability was assessed using two main gap metrics. The first metric measures the difference between cross-validation performance and private leaderboard performance:

$$\text{CV-private gap} = \text{CV AUC} - \text{Private AUC}$$

The second metric measures the difference between public leaderboard performance and private leaderboard performance:

$$\text{Public-private gap} = \text{Public AUC} - \text{Private AUC}$$

The private leaderboard was treated as the closest available proxy for hidden generalisation performance because it is based on the final unseen evaluation split. Smaller absolute gaps indicate stronger agreement between validation estimates and private leaderboard outcomes.

The study also computed Pearson and Spearman correlations between cross-validation AUC and private

leaderboard AUC, and between public leaderboard AUC and private leaderboard AUC. Correlations were computed overall, by competition, by model family, and using a within-competition centred approach. The centred analysis was included because overall correlations can be inflated when competitions have different baseline score ranges.

H. Bayesian Melding Framework

Bayesian melding was used to integrate cross-validation, public leaderboard, and private leaderboard scores as noisy observations of an underlying latent model performance parameter. For each model m , the latent true generalisation performance is denoted by θ_m .

For each model, the observed evaluation signals are defined as:

$$\mathbf{x}_m = \{\text{CV}_m, \text{PLB}_m, \text{PrLB}_m\}$$

where CV_m is the cross-validation AUC, PLB_m is the public leaderboard AUC, and PrLB_m is the private leaderboard AUC.

The Bayesian melding objective is to estimate the posterior distribution of latent true model performance given the observed evaluation signals:

$$p(\theta_m | \text{CV}_m, \text{PLB}_m, \text{PrLB}_m)$$

A weak Gaussian prior was placed over the latent model performance:

$$\theta_m \sim N(\mu_0, \tau_0^2)$$

The observation model is defined as:

$$\text{CV}_m = \theta_m + b\text{CV} + \varepsilon\text{CV}$$

$$\text{PLB}_m = \theta_m + b\text{PLB} + \varepsilon\text{PLB}$$

$$\text{PrLB}_m = \theta_m + \varepsilon\text{Pr}$$

where $b\text{CV}$ is the systematic bias of cross-validation, $b\text{PLB}$ is the systematic bias of the public leaderboard, and ε terms represent noise.

Equivalently, the likelihood terms can be written as:

$$p(CV_m | \theta_m) = N(\theta_m + bCV, \sigma CV^2)$$

$$p(PLB_m | \theta_m) = N(\theta_m + bPLB, \sigma PLB^2)$$

$$p(PrLB_m | \theta_m) = N(\theta_m, \sigma Pr^2)$$

The posterior is therefore proportional to the prior multiplied by the three likelihood terms:

$$p(\theta_m | CV_m, PLB_m, PrLB_m) \propto p(\theta_m) \cdot p(CV_m | \theta_m) \cdot p(PLB_m | \theta_m) \cdot p(PrLB_m | \theta_m)$$

The private leaderboard was treated as the closest observed proxy to true generalisation performance. Cross-validation and public leaderboard biases were estimated empirically as:

$$bCV = \text{mean}(CV \text{ AUC} - \text{Private AUC})$$

$$bPLB = \text{mean}(\text{Public AUC} - \text{Private AUC})$$

Noise terms were estimated using the standard deviations of the corresponding gaps:

$$\sigma CV = \text{std}(CV \text{ AUC} - \text{Private AUC})$$

$$\sigma PLB = \text{std}(\text{Public AUC} - \text{Private AUC})$$

The private leaderboard uncertainty was assigned as half of the smaller estimated noise term, reflecting its role as the closest observed proxy to true generalisation while still acknowledging that it is a finite test-set estimate rather than a perfect truth.

After bias correction, the three observed signals were combined using inverse-variance weighting. The posterior estimate of latent true performance was computed as:

$$\theta_m^* = (wCV \cdot CV_{\text{corrected}} + wPLB \cdot Public_{\text{corrected}} + wPr \cdot Private) / (wCV + wPLB + wPr)$$

where each weight is the inverse of the corresponding variance:

$$w = 1 / \sigma^2$$

This produces a posterior true performance estimate for each leaderboard-scored model, along with posterior

uncertainty. The Bayesian melding output was used to compare the fused estimate against private leaderboard performance and to summarise model reliability across competitions and model families.

I. Ethical and Reproducibility Considerations

The study used publicly available completed Kaggle competitions. No private personal data was collected by the researcher, and no attempt was made to identify individuals represented in the datasets. The research focused only on model evaluation behaviour and leaderboard reliability.

All experiments used consistent preprocessing, model generation, and validation procedures across competitions. Random seeds were controlled during model configuration generation. Final datasets were saved as CSV files to support reproducibility, including the full 3,000-model cross-validation population, the 300-model leaderboard-scored dataset, and the Bayesian melding output.

V. RESULTS

This section presents the empirical results from the full 3,000-model cross-validation population and the 300-model leaderboard-scored subset. The results are organised into four parts: full cross-validation population behaviour, leaderboard-scored subset performance, validation reliability analysis, and Bayesian melding outputs.

A. Full 3,000-Model Cross-Validation Population

The full experimental population consisted of 3,000 trained model configurations, with 1,000 models evaluated for each of the three Kaggle competitions. This population was used to characterise the broader model search space before analysing the leaderboard-scored subset.

Table I summarises the distribution of model families in the full cross-validation population. LightGBM and XGBoost formed the largest groups, with 1,050 models each across the three competitions. CatBoost contributed 450 models, Logistic Regression contributed 225 models, and the bagging-based tree models contributed 126 ExtraTrees models and 99 Random Forest models.

TABLE I. Full Cross-Validation Population by Model Family

Model family	Number of models
LightGBM	1050
XGBoost	1050
CatBoost	450
Logistic Regression	225
ExtraTrees	126
Random Forest	99
Total	3000

The cross-validation distributions varied strongly across competitions. Homesite had the highest overall AUC range, with boosting models reaching approximately 0.966. Santander achieved private-leaderboard-level performance around the mid-to-high 0.80s, while Porto Seguro was lower in absolute AUC after converting Normalized Gini to AUC. These differences reflect dataset-specific difficulty and baseline score ranges rather than direct differences in modelling quality.

TABLE II. Full Cross-Validation Population Summary by Competition and Model Family

Competition	Model family	Models	Mean CV AUC	Min CV AUC	Max CV AUC
Homesite	LightGBM	350	0.9634	0.9440	0.9661
Homesite	XGBoost	350	0.9631	0.9437	0.9660
Homesite	CatBoost	150	0.9624	0.9520	0.9660
Homesite	Random Forest	33	0.9514	0.9455	0.9580
Homesite	ExtraTrees	42	0.9507	0.9417	0.9563
Homesite	Logistic Regression	75	0.9413	0.9350	0.9428
Porto	LightGBM	350	0.6363	0.6049	0.6413
Porto	XGBoost	350	0.6362	0.6009	0.6420
Porto	CatBoost	150	0.6360	0.6225	0.6408
Porto	Logistic Regression	75	0.6195	0.6151	0.6204
Porto	ExtraTrees	42	0.6188	0.5982	0.6264
Porto	Random Forest	33	0.6135	0.5980	0.6237
Santander	XGBoost	350	0.8728	0.7862	0.8947
Santander	LightGBM	350	0.8728	0.7854	0.8941
Santander	CatBoost	150	0.8664	0.7953	0.8951
Santander	ExtraTrees	42	0.8622	0.8334	0.8773
Santander	Logistic Regression	75	0.8573	0.8355	0.8595
Santander	Random Forest	33	0.8514	0.8411	0.8629

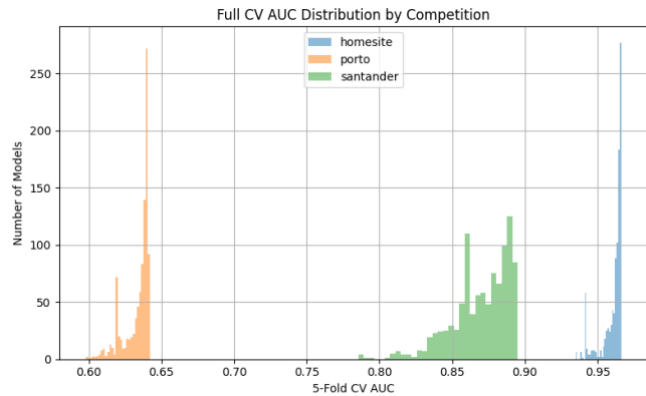


Fig. 2. Distribution of 5-fold cross-validation AUC across the full 1,000-model population for each competition.

Across the full population, gradient boosting methods generally achieved the strongest cross-validation scores. LightGBM, XGBoost, and CatBoost were consistently competitive across all three competitions. Logistic Regression and bagging-based models generally performed lower, although their inclusion was important because the objective was to study validation reliability across a wide range of model strengths rather than only high-performing models.

B. Leaderboard-Scored Subset

The leaderboard-scored dataset contained 300 models, with 100 models per competition. This subset was selected from the full 3,000-model population across model families and low, median, and high cross-validation performance ranges. The final model-family distribution is shown in Table III.

TABLE III. Leaderboard-Scored Dataset by Model Family

Model family	Number of models
LightGBM	75
XGBoost	75
CatBoost	60
Logistic Regression	45
ExtraTrees	23
Random Forest	22
Total	300

Table IV summarises the leaderboard-scored subset by competition. Homesite achieved the highest mean private AUC, followed by Santander and Porto. The absolute AUC values are not directly comparable as measures of dataset difficulty because each competition has different data distributions and target definitions. However, within each competition, the relationship between cross-validation, public leaderboard, and private leaderboard performance is directly relevant to validation reliability.

TABLE IV. Leaderboard-Scored Summary by Competition

Competition	Models	Mean CV AUC	Mean Public AUC	Mean Private AUC	Mean Abs. CV-Private Gap	Mean Abs. Public-Private Gap
Homesite	100	0.955840	0.956981	0.956401	0.000573	0.000585
Porto	100	0.626901	0.625845	0.627521	0.000999	0.001924
Santander	100	0.858434	0.858785	0.855992	0.003576	0.002799

Homesite showed the smallest validation discrepancy, with a mean absolute CV-private gap of 0.000573 AUC. Porto also showed a small gap of 0.000999 AUC. Santander had the largest discrepancy, with a mean absolute CV-private gap of 0.003576 AUC. This suggests that cross-validation remained reliable overall, but its reliability varied by competition.

C. Correlation Between Validation and Private Leaderboard Performance

The primary research question asks whether cross-validation AUC reliably predicts private leaderboard AUC. Table V reports Pearson and Spearman correlations between cross-validation AUC and private leaderboard AUC, as well as between public leaderboard AUC and private leaderboard AUC.

TABLE V. Correlation Between Evaluation Signals

Group	Models	CV-Private Pearson	CV-Private Spearman	Public-Private Pearson	Public-Private Spearman
Overall	300	0.999849	0.998848	0.999919	0.999245
Homesite	100	0.999342	0.987109	0.999481	0.995439
Porto	100	0.996967	0.989533	0.997015	0.992010
Santander	100	0.995336	0.992252	0.999158	0.992158
Within-competition centred	300	0.995804	0.994294	0.998821	0.996460

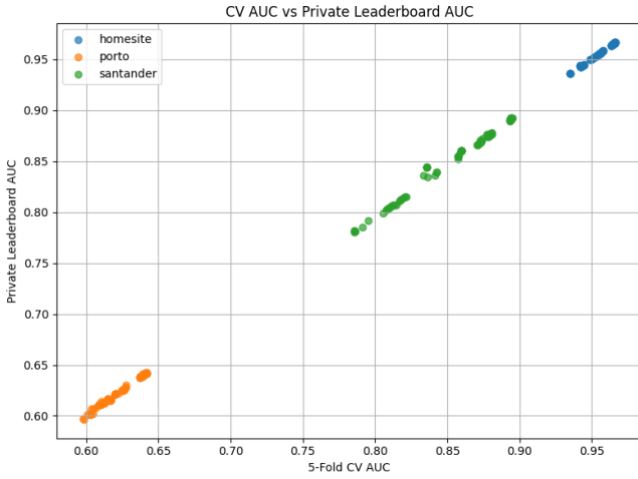


Fig. 3. Relationship between cross-validation AUC and private leaderboard AUC across the 300 leaderboard-scored models.

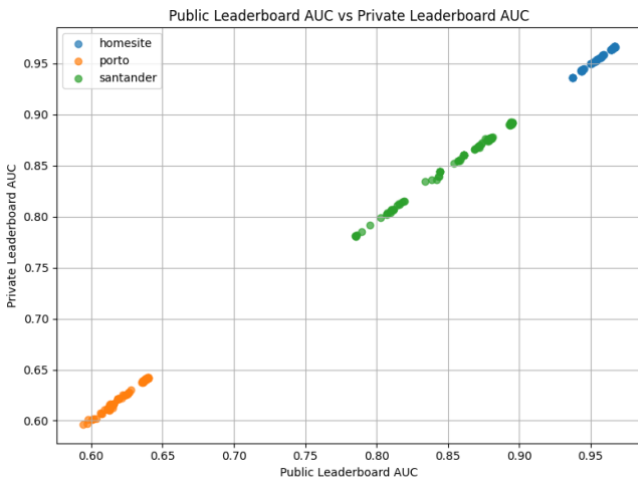


Fig. 4. Relationship between public leaderboard AUC and private leaderboard AUC across the 300 leaderboard-scored models.

The correlations are very high across all competitions. The overall CV-private Pearson correlation was 0.999849, while the public-private Pearson correlation was 0.999919. However, the overall result is partly influenced by the

different baseline score ranges across competitions. To address this, a within-competition centred analysis was also conducted. After centring scores within each competition, CV-private Pearson correlation remained very high at 0.995804, and public-private Pearson correlation remained 0.998821. This confirms that the strong relationship is not only an artefact of combining competitions with different score scales.

Public leaderboard scores were slightly more correlated with private leaderboard scores than cross-validation scores. This is expected because the public and private leaderboards are both based on held-out Kaggle test data, whereas cross-validation uses only the training data. However, the difference was small, suggesting that stratified 5-fold cross-validation was still highly reliable in these experiments.

D. Validation Gap Analysis

The validation gap analysis measures whether cross-validation and public leaderboard scores systematically overestimate or underestimate private leaderboard performance. A positive CV-private gap means cross-validation is optimistic relative to the private leaderboard. A negative gap means cross-validation is pessimistic.

TABLE VI. Gap Summary by Competition

Competition	Mean CV-Private Gap	Std CV-Private Gap	Mean Abs. CV-Private Gap	Mean Public-Private Gap	Mean Abs. Public-Private Gap
Homesite	-0.000562	0.000392	0.000573	0.000580	0.000585
Porto	-0.000619	0.001075	0.000999	-0.001676	0.001924
Santander	0.002442	0.003249	0.003576	0.002793	0.002799

Homesite and Porto showed slightly negative mean CV-private gaps, indicating that cross-validation was mildly pessimistic on average. Santander showed a positive mean CV-private gap of 0.002442 AUC, indicating mild optimism. Santander also had the largest average absolute CV-private gap, suggesting that validation reliability was weaker for Santander than for Homesite or Porto.

Public-private gaps were also small. Homesite showed almost no discrepancy between public and private leaderboard scores. Porto showed a negative public-private gap, meaning public leaderboard scores were slightly lower than private leaderboard scores. Santander showed a positive public-private gap, meaning public leaderboard scores were mildly optimistic relative to private leaderboard scores.

TABLE VII. Gap Summary by Model Family

Model family	Models	Mean CV-Private Gap	Mean Abs. CV-Private Gap	Mean Public-Private Gap	Mean Abs. Public-Private Gap
Random Forest	22	0.000836	0.001596	0.000653	0.001622
XGBoost	75	0.000847	0.001656	0.000944	0.001935
CatBoost	60	0.000981	0.001685	0.000483	0.001858

LightGBM	75	0.000710	0.001687	0.000869	0.002038
ExtraTrees	23	0.000778	0.001832	0.000173	0.001483
Logistic Regression	45	-0.001908	0.001908	-0.000301	0.001143

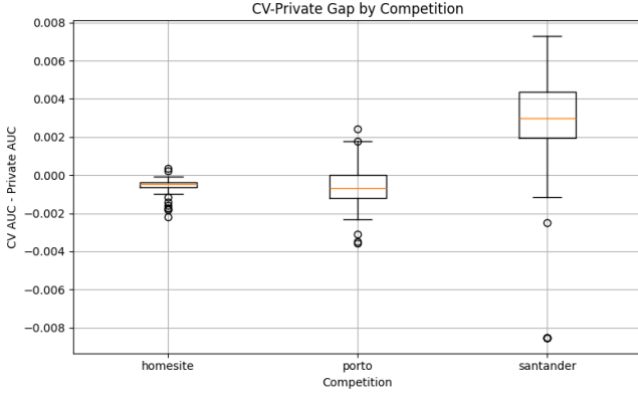


Fig. 5. Distribution of CV-private leaderboard gaps by competition.

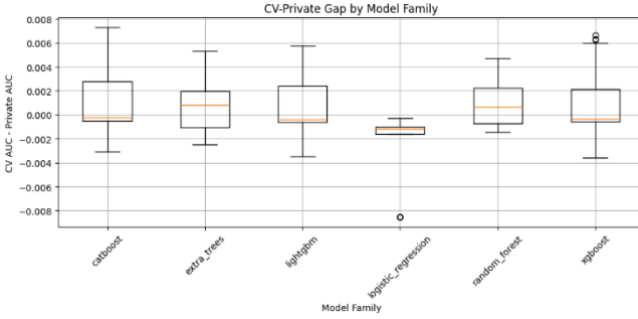


Fig. 6. Distribution of CV-private leaderboard gaps by model family.

At the model-family level, absolute CV-private gaps were small across all families. Random Forest had the lowest mean absolute CV-private gap at 0.001596, followed closely by XGBoost, CatBoost, and LightGBM. Logistic Regression had a negative mean CV-private gap, indicating that it was generally pessimistic relative to private leaderboard performance. ExtraTrees and Logistic Regression showed slightly larger absolute CV-private gaps than the strongest boosting families, but the differences were still small in practical terms.

E. Bayesian Melding Results

Bayesian melding was applied to integrate cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. The estimated global CV bias was 0.000420 AUC, while the estimated public leaderboard bias was 0.000566 AUC. These values are close to zero, suggesting that both cross-validation and public leaderboard scores were only mildly optimistic on average.

The estimated noise term for cross-validation was $\sigma_{CV} = 0.002446$, while the estimated noise term for the public leaderboard was $\sigma_{Public} = 0.002138$. The public leaderboard therefore had slightly lower estimated noise than cross-validation, which is consistent with the very high public-private correlations observed earlier. The private leaderboard

uncertainty was assigned as $\sigma_{Private} = 0.001069$, reflecting its role as the closest observed proxy for true generalisation.

TABLE VIII. Bayesian Melding Bias and Noise Estimates

Parameter	Estimate
CV bias	0.000420
Public leaderboard bias	0.000566
σ_{CV}	0.002446
σ_{Public}	0.002138
$\sigma_{Private}$	0.001069

Table IX summarises the posterior true performance estimates by competition. The posterior mean remained close to the private leaderboard mean for each competition, indicating that the Bayesian melding procedure produced fused estimates consistent with the strongest available proxy for generalisation.

TABLE IX. Bayesian Posterior Summary by Competition

Competition	Models	Posterior Mean	Posterior Std	Mean Posterior Uncertainty	Mean Private AUC	Mean CV AUC	Mean Public AUC
Homesite	100	0.956274	0.009911	0.000890	0.956401	0.955840	0.956981
Porto	100	0.626994	0.013658	0.000890	0.627521	0.626901	0.625845
Santander	100	0.856646	0.033072	0.000890	0.855992	0.858434	0.858785

Homesite had the highest posterior mean performance, followed by Santander and Porto. This ranking reflects the different AUC ranges of the competitions rather than a direct comparison of dataset difficulty. More importantly, the posterior means were close to the corresponding private leaderboard means, showing that the fused estimates did not drift away from the final evaluation target.

TABLE X. Bayesian Posterior Summary by Model Family

Model family	Models	Posterior Mean	Posterior Std	Mean Private AUC	Mean CV AUC	Mean Public AUC	Mean Abs. CV-Private Gap
Random Forest	22	0.830430	0.146991	0.830360	0.831196	0.831013	0.001596
CatBoost	60	0.818171	0.139065	0.818111	0.819092	0.818594	0.001685
XGBoost	75	0.816234	0.141208	0.816112	0.816958	0.817055	0.001656
LightGBM	75	0.815812	0.140976	0.815721	0.816431	0.816590	0.001687
Logistic Regression	45	0.804753	0.137481	0.805212	0.803305	0.804911	0.001908
ExtraTrees	23	0.783232	0.140422	0.783253	0.784031	0.783426	0.001832

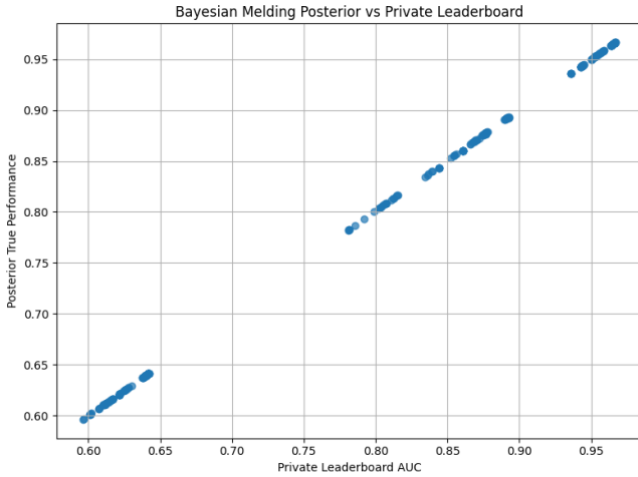


Fig. 7. Bayesian melding posterior true performance estimates compared with private leaderboard AUC.

At the model-family level, posterior mean performance was highest for Random Forest, followed by CatBoost, XGBoost, and LightGBM. However, this ordering should be interpreted carefully because model-family averages combine results across competitions with different baseline score ranges and because the leaderboard subset was selected for validation coverage rather than leaderboard ranking. The more important result is that posterior estimates remained close to private leaderboard means across all model families.

VI. DISCUSSION

The results indicate that stratified 5-fold cross-validation was highly reliable in the three Kaggle tabular binary classification competitions studied. Across the 300 leaderboard-scored models, cross-validation AUC was strongly correlated with private leaderboard AUC. This relationship remained strong even after scores were centred within each competition, which shows that the finding was not simply caused by combining competitions with different baseline AUC ranges.

The small mean CV-private gap suggests that cross-validation did not systematically misestimate private leaderboard performance in a large way. The global estimated CV bias was only 0.000420 AUC, which is practically small. However, the direction and magnitude of the gap differed by competition. Homesite and Porto showed slightly pessimistic cross-validation estimates, while Santander showed mild optimism. This suggests that validation reliability is not universal; it depends on the dataset, hidden test split, model family, and competition design.

Public leaderboard scores were slightly more aligned with private leaderboard scores than cross-validation scores. This is expected because both public and private leaderboard scores are based on Kaggle hidden test data. However, the difference between public-private and CV-private reliability was small. This finding is important because it suggests that a carefully designed local cross-validation strategy can approximate hidden leaderboard behaviour closely, even without relying heavily on public leaderboard feedback.

The results also show that validation reliability was broadly consistent across model families. Gradient boosting models such as LightGBM, XGBoost, and CatBoost achieved strong cross-validation and leaderboard performance, but their validation gaps were not dramatically different from simpler models. Logistic Regression showed a slightly negative CV-private gap, indicating that it was more pessimistic on average. This may be because linear models are less flexible and less likely to overfit the validation folds compared with complex ensemble models.

Bayesian melding provided a useful framework for formalising the relationship between validation sources. By treating cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance, the framework allowed systematic bias and noise to be estimated explicitly. The posterior true performance estimates remained close to private leaderboard means, showing that the fused estimates were well-calibrated to the final evaluation target.

The novelty of this research lies in combining large-scale empirical model generation with Bayesian validation modelling. Rather than comparing only a few manually tuned models, the study trained 3,000 configurations across three competitions and collected leaderboard scores for a representative subset of 300 models. This design provides a stronger empirical basis for assessing validation reliability than a small number of isolated model comparisons.

The findings are consistent with prior work showing that leaderboard overfitting is not always severe in Kaggle competitions, especially when evaluation splits are sufficiently stable. However, the results also support prior warnings about validation bias and adaptive evaluation. Even though gaps were small in this study, they were not zero, and Santander showed larger discrepancies than the other competitions. Practitioners should therefore avoid treating cross-validation or public leaderboard scores as perfect measures of generalisation.

A. Limitations

This study has several limitations. First, leaderboard scores were collected for a representative subset of 100 models per competition rather than for all 1,000 trained models per competition. This was due to Kaggle submission limits and rate throttling. Although the subset was selected across model families and performance ranges, it remains a sample of the full model population.

Second, the study focused only on completed Kaggle tabular binary classification competitions. The findings may not generalise to multiclass classification, regression, time series, image, or natural language competitions. Third, only stratified 5-fold cross-validation was used as the main validation strategy. Repeated cross-validation or nested cross-validation could provide additional insight into validation stability, but would require substantially more computation.

Fourth, the private leaderboard was treated as the closest available proxy for true generalisation performance. However, the private leaderboard is still a finite hidden test split and may contain sampling noise. It should therefore be interpreted as an operational proxy for true generalisation rather than a perfect measurement.

Finally, the Bayesian melding model used a simple Gaussian bias-and-noise formulation. This was appropriate for interpretability and alignment with the research question, but more complex hierarchical Bayesian models could be explored in future work.

B. Future Work

Future work could extend this research in several directions. First, the experiment could be repeated across more Kaggle competitions to test whether the observed reliability patterns hold across different domains and dataset sizes. Second, future studies could compare multiple validation strategies, such as repeated cross-validation, nested cross-validation, adversarial validation, and time-based validation splits.

Third, more advanced Bayesian models could be used. A hierarchical Bayesian model could estimate competition-specific and model-family-specific bias terms rather than using only global bias estimates. This would allow the model to capture differences between competitions such as Santander, Porto, and Homesite more explicitly.

Fourth, future work could use the 1,000-model cross-validation population to estimate leaderboard outcomes for unsubmitted models. This was not performed in the present study to avoid adding unnecessary modelling assumptions close to the final analysis, but it remains a possible extension.

Finally, future work could analyse how leaderboard submission behaviour affects validation reliability. For example, the number of submissions, the order of submissions, and adaptive model changes based on public leaderboard feedback may influence the degree of public leaderboard overfitting.

VII. CONCLUSION

This study evaluated the reliability of cross-validation in Kaggle tabular binary classification competitions using a large-scale empirical experiment and Bayesian melding framework. Three completed Kaggle competitions were analysed: Santander Customer Transaction Prediction, Porto Seguro's Safe Driver Prediction, and Homesite Quote Conversion. Across these competitions, 3,000 model configurations were trained and evaluated using stratified 5-fold cross-validation. A representative subset of 300 models was then scored on Kaggle public and private leaderboards.

The results show that cross-validation AUC was a highly reliable predictor of private leaderboard AUC in the studied competitions. The within-competition centred Pearson correlation between cross-validation AUC and private

leaderboard AUC was 0.9958, while the centred Pearson correlation between public and private leaderboard AUC was 0.9988. The mean global cross-validation bias was also very small at 0.00042 AUC, suggesting that stratified 5-fold cross-validation provided a well-calibrated estimate of hidden leaderboard performance.

Competition-level differences were still observed. Homesite showed the smallest validation gap, Porto showed slightly pessimistic cross-validation and public leaderboard behaviour, and Santander showed mild optimism in both cross-validation and public leaderboard scores. This indicates that cross-validation reliability is strong overall but still depends on dataset characteristics and competition design.

Bayesian melding provided a useful framework for integrating cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. The posterior estimates remained close to private leaderboard performance, supporting the use of Bayesian melding as an interpretable validation reliability framework.

Overall, the study suggests that carefully implemented stratified cross-validation can be highly reliable in Kaggle tabular binary classification competitions. However, validation estimates should still be interpreted with uncertainty, especially when public leaderboard feedback, model selection, and dataset-specific effects are involved.

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